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KINEMATIC MODELING OF 6 - DOF ROBOTIC MANIPULATOR WITH A PRISMATIC JOINT

Abstract

The investigation of the movement of a robot manipulator is one of the most important parts of the design process. This research involves the design and analysis of a six-DOF manipulator with a prismatic joint. However, the design and analysis of a robot involve modelling its forward and inverse kinematics. The forward and inverse kinematics are modeled using D-H parameters. The proposed model makes it possible to control the manipulator to achieve any reachable position and orientation in an unstructured environment. This study focuses on the development of a Graphical User Interface Development Environment (GUIDE) in MATLAB that can be used for the purposes of forward and inverse kinematics.

Keywords: robot manipulator, Denavit-Hartenberg (D-H), forward kinematics, inverse kinematics, MATLAB

Introduction

The robotic equipment has several uses in various industries and daily life, including tasks such as painting, pick-and-place operations, welding, grasping, assembling, material removal, medical procedures, and more. In the manufacturing industry, robotic manipulators are the most used robots. Manipulators are implemented in several sectors with the aim of significantly enhancing both productivity and product quality. Studying the kinematic behaviour of a robot is an essential aspect of modelling it.

A kinematic model focuses on the movement of the robot, disregarding the forces that cause the movement. The resulting inverse kinematics algorithm can provide an accurate solution for every attainable position and orientation of the end-effector. Inverse kinematics has many positions and was difficult to solve as degree of freedom increased [1].

Robot kinematics

Robot kinematics is a field of research that focuses on the analysis and description of the motion exhibited by kinematic chains with multiple degrees of freedom, which constitute the foundational structure of robotic systems [2]. Figure 1 shows the relationship between the joint angles and the position of the end-effector using the techniques of forward and inverse kinematics. The technique of forward kinematics involves utilising the kinematic equations of a robotic system to determine the precise position of the end-effector based on defined values for the joint parameters. The technique of inverse kinematics involves determining the joint angles required to achieve a desired end-effector position. The robot's dimensions and kinematics equations establish the boundaries of the space that the robot can access, which is commonly referred to as its workspace [3].

The manipulator of 6 DOF was built utilising the SolidWorks software, as shown in Figure 2. This manipulator has five links (H, L_1, L_2, L_3, L_4), five angles (q_1, q_2, q_3, q_4, q_5) and one prismatic joint (q_6).

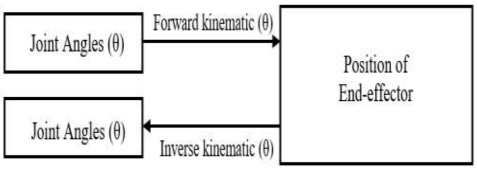

Figure 1 – Relation between joint angles & end – effector position

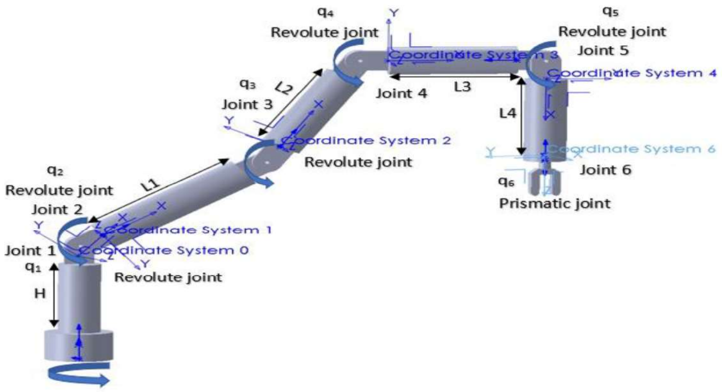

Figure 2 – The structure of the 6-DOF manipulator

Denavit-Hartenberg method

The most common frame of reference selection convention in robotics is Denavit-Hartenberg (D-H) parameters. The four parameters a_i, α_i, d_i, and θ_i are referred to as link length, link twist, joint offset, and joint angle, respectively. In Table 1, the manipulator's structural parameters were set out in detail. The representation of each homogeneous transformation T_i is expressed as the result of four fundamental transformations [4].

Table 1 – The Denavit – Hartenberg parameters

Joint no.	Joint angle θ_i	Joint offset d_i	Link length a_i	Twist angle α_i
0-1 Base	q_1	H	0	0
1-2 Shoulder	q_2	0	0	90º
2-3 Elbow	q_3	0	L_1	0
3-4 Elbow	q_4	0	L_2	0
4-5 Wrist	q_5	0	L_3	0
5-6 Gripper	q_6	L_4	0	90º

The process of determining the transformations between two adjacent joints can be achieved by replacing the parameters from the parameters table into the transformation matrix, utilising equation (1).

$$T_i^{i-1} = \begin{bmatrix} c_{\theta i} & -s_{\theta i}c_{\alpha i} & s_{\theta i}\, s_{\alpha i} & a_i\, c_{\theta i} \\ s_{\theta i} & c_{\theta i}c_{\alpha i} & -c_{\theta i}s_{\alpha i} & a_i\, s_{\theta i} \\ 0 & s_{\alpha i} & c_{\alpha i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1}$$

The order of transformations follows a consecutive sequence, beginning from the first joint and continuing to the nth joint. Equation (2) represents the total transformation up to the end-effector.

$$T_i^0 = T_1^0 . T_2^1 . T_3^2 .. T_i^{i-1} = \begin{bmatrix} R_n^0 & P_n^0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

The total transformation T_6^0:

$$T_6^0 = \begin{bmatrix} c_{12}\, c_{3456} & s_{12} & c_{12}\, s_{3456} & L_3\, c_{12}\, c_{345} + L_4\, s_{12} + \; c_{12}\,(L_2\, s_{34} + L_1\, c_3) \\ s_{12}\, c_{3456} & -\, c_{12} & s_{12}\, s_{3456} & L_3\, s_{12}\, c_{345} - L_4\, c_{12} + \; s_{12}\,(L_2\, c_{34} + L_1\, s_3) \\ s_{3456} & 0 & -\, c_{3456} & L_3\, s_{345} + L_2\, s_3 + H \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

When working with inverse kinematics, we use the total transformation matrix from equation (2) and multiply it by the inverse of the initial transformation matrix. This allows us to calculate the joint angles by comparing the matrix elements in equation (3).

$$(T_{\mathrm{n}}^0)^{-1} \times T_{\mathrm{n}}^0 \;=\; T_2^1 \ldots T_{\mathrm{n}}^{\mathrm{n}-1} \tag{3}$$

$$\text{where: } (T_{\mathrm{n}}^0)^{-1} = \frac{Adj\;(T_{\mathrm{n}}^0)}{Det\;(T_{\mathrm{n}}^0)}$$

The 6 - DOF manipulator coordinate systems are established using the D-H method [3, 4]. The coordinate frames for each link were determined systematically, beginning at the base and ending at the end effector, in order to conduct a kinematic analysis. The representation of links and frame assignment for the manipulator is shown in Figure 3.

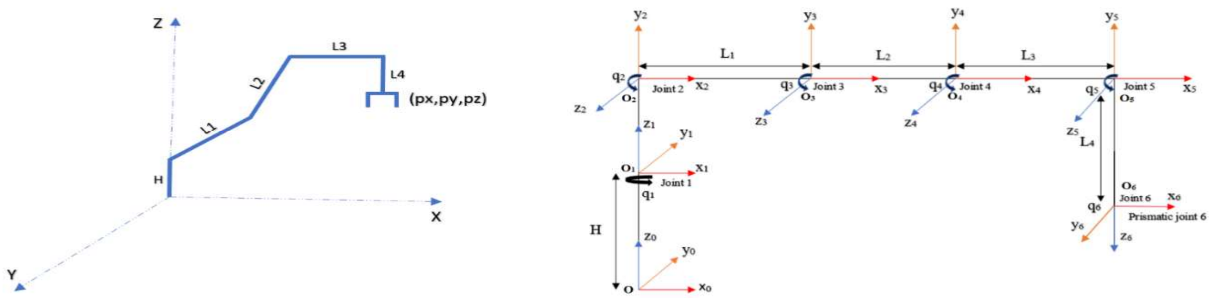

Figure 3 – The coordinate systems of the 6-DOF manipulator

Algorithm

This study used MATLAB and the robotic toolbox developed as software tools to execute kinematic analysis. Figure 4 displays the flowchart illustrating the algorithm employed for the preparation of the MATLAB GUIDE utilised in kinematic analysis. In this process, the initial step involves the user providing the D-H parameters, which consist of the length, link twist, and joint angle. After that, if the user intends to perform forward kinematics, the joint angles are provided as input. Figure 5 displays a graphic representation of the MATLAB GUIDE interface. Figure 6 displays a graphic representation of the RRRRRP manipulator in its original configuration. Figure 7 displays a screenshot representing the complete configuration of a RRRRRP manipulator after the computation of forward kinematics.

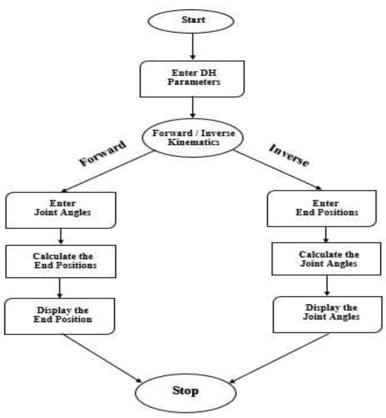

Figure 4 – Flow chart algorithm

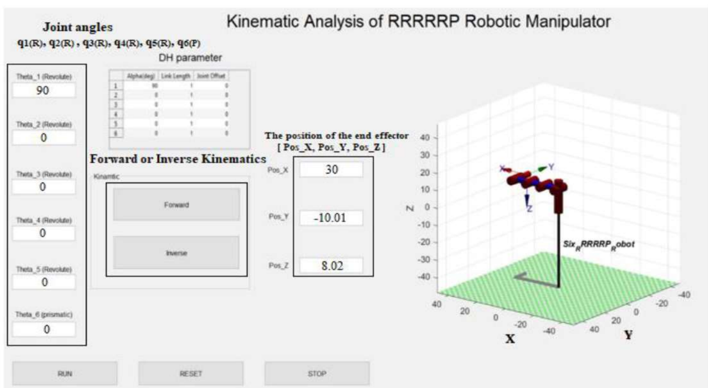

Figure 5 – MATLAB guide

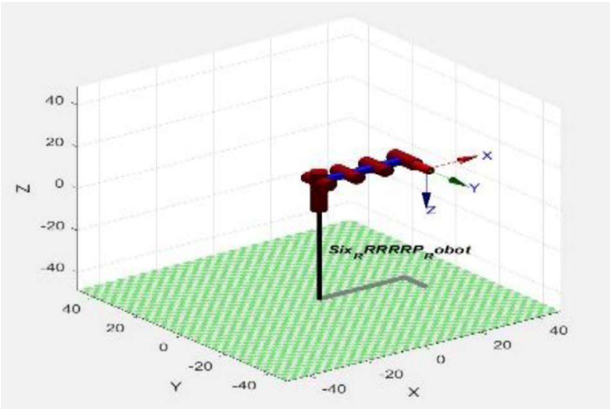

Figure 6 – Screenshot of the initial position of the RRRRRP manipulator

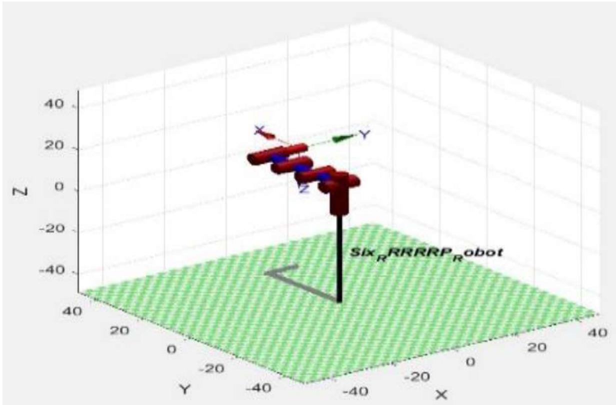

Figure 7 – Screenshot of the final position after forward kinematics

Results and discussion

For the particular D-H parameters presented in Figure 6, each of the forward and inverse kinematics was successfully computed. Tables 2 and 3 display the expected and obtained findings for forward and inverse kinematics, respectively. After getting the obtained results, it can be seen that percentage errors are almost negligible because the difference in is very small.

Table 2 – Forward Kinematics Results

θ_1	90	Expected	X	30
θ_2	0		Y	-10
θ_3	0		Z	8
θ_4	0	Obtained	X	30
θ_5	0		Y	-10.01
θ_6	0		Z	8.02

Table 3 – Inverse Kinematics Results

		Expected		Obtained	
X	10	θ_1	90	θ_1	90
		θ_2	0	θ_2	-0.00707
Y	30	θ_3	0	θ_3	0.0094
		θ_4	0	θ_4	-0.0028
Z	8	θ_5	0	θ_5	-0.0056
		θ_6	0	θ_6	-0.0014

Conclusions

The research aims to simplify the complex calculations of forward and inverse kinematics in the robotic manipulator control field. It is apparent that the MATLAB GUIDE is simple to use and, at the same time, delivers valid and accurate findings. With the MATLAB guide, numerous end positions can be generated by inputting various joint angles through forward kinematics. Additionally, various joint angles can be calculated by inputting the different end positions through inverse kinematics. For future studies, dynamic modelling can be executed with MATLAB.

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КИНЕМАТИЧЕСКИЙ АНАЛИЗ РОБОТИЗИРОВАННЫХ МАНИПУЛЯТОРОВ С ШЕСТЬЮ СТЕПЕНЯМИ СВОБОДЫ И ПРИЗМАТИЧЕСКИМ СОЧЛЕНЕНИЕМ

Аннотация

Исследование движения робота-манипулятора является одной из важнейших частей процесса проектирования. Статья включает в себя проектирование и анализ 6 - DOF манипулятора с призматическим сочленением. Однако проектирование и анализ робота предполагают моделирование его прямой и обратной кинематики. Мы смоделировали прямую и обратную кинематику, используя параметры D-H. Предложенная модель позволяет управлять манипулятором для достижения любого положения и ориентации в неорганизованной среде. Статья посвящена разработке среды Interface Development Environment (GUIDE) в MATLAB, которая может быть использована для целей прямой и обратной кинематики.

Ключевые слова: робот-манипулятор, Денавит-Хартенберг (D-H), прямая кинематика, обратная кинематика, MATLAB